



3 Complex Analysis 2024

Tutorial 3

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Mean value theorem

Problem 11 Consider the function $f(z) = \frac{1}{z}$ which is holomorphic in $\mathbb{C} \setminus \{0\}$. Take $z_0 = 1$, $z_1 = 1 + i$.

- (a) Calculate $\frac{f(z_1) - f(z_0)}{z_1 - z_0}$.
- (b) Show that there is no $c \in [z_0, z_1]$ with $f'(c) = \frac{f(z_1) - f(z_0)}{z_1 - z_0}$.

Series

Problem 12 Prove that if $\sum_{k=1}^{\infty} a_k$ is a convergent series of complex numbers, then $\lim_{k \rightarrow \infty} a_k = 0$.

Problem 13

- (a) Prove the Comparison Test for complex series: Let $\sum_{k=1}^{\infty} a_k$ be a complex series. Assume that for each $k \in \mathbb{N}$ there is a real number $c_k \geq 0$ such that $|a_k| \leq c_k$. If $\sum_{k=1}^{\infty} c_k < \infty$, then $\sum_{k=1}^{\infty} a_k$ converges.
- (b) Prove that if a complex series $\sum_{k=1}^{\infty} a_k$ converges absolutely, then it converges.

Power series

Problem 14 Use the Weierstrass M-Test to show that the series $\sum_{k=1}^{\infty} \frac{z^k}{k^k}$ converges uniformly on any disc of the form $\{z \in \mathbb{C} : |z| \leq r\}$, $r > 0$.

Problem 15 We know that the series $\sum_{k=0}^{\infty} z^k$ converges uniformly on any disc of the form $\{z \in \mathbb{C} : |z| \leq r\}$, $0 < r < 1$.

- (a) Show this directly using the definition of uniform convergence.
- (b) Show that the series does not converge uniformly on the disc $\{z \in \mathbb{C} : |z| < 1\}$.